

Dihedral 2

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Problem 1. *Exhibit $2n$ elements of D_n – n rotations and n -reflections. Note the difference for reflections in case n is even or odd*

Proof. Assume the polygon's vertices are placed on the n th roots of unity.

$\rho_\theta = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$ is a rotation around the origin by a fixed angle θ . r_L is a reflection about the line L through the origin. Define $r_{a,b}$ be the reflection across the line through points a, b . By definition $\forall \theta \in \mathbb{R} \rho_\theta, r_L \in O(2)$.

The rotations in D_n are $\rho_{2k\pi/n} \forall k \in [0, 1, \dots, n]$. The reflections in D_n are a bit more complicated, and depends on whether n is even or odd. Since the reflections in D_n must flip the n -gon onto itself, it means the line must split the n -gon into equal areas. If n is even, these lines will pass through 2 vertices $e^{\frac{2ik\pi}{n}}$ and $-e^{\frac{2ik\pi}{n}}$ and the origin or through only the origin and in between vertices at $e^{\frac{2ik\pi}{n}}$ and $e^{\frac{2ik+1\pi}{n}}$, thus through point $e^{\frac{2i(k+1/2)\pi}{n}}$, assuming the vertices of the n -gon are placed on the roots of unity. Note lines through $e^{\frac{2ik\pi}{n}}$ and $-e^{\frac{2ik\pi}{n}}$ passing through the origin and lines passing through $-e^{\frac{2ik\pi}{n}}, 0, e^{\frac{2ik\pi}{n}}$ are the same. Thus we only have $n/2$ lines for even n and for lines passing through 2 vertices of the n -gon. Furthermore lines passing through $e^{\frac{2i(k+1/2)\pi}{n}}, 0$ are the same as $-e^{\frac{2i(k+1/2)\pi}{n}} = e^{\frac{\pi(2k+n+1)}{n}}$ and 0, thus there are only $n/2$ choices for k . Thus the reflections if n is even are $\{r_{e^{\frac{2ik\pi}{n}}, 0} \mid k \in [0, \dots, n-1]\}$. If n is odd, the lines of reflections can only pass through the origin and one vertex. Since there are n vertices there must be n lines of reflection. The reflections in D_n are $\{r_{e^{\frac{2ik\pi}{n}}, 0} \mid k \in [0, \dots, n-1]\}$ $\square$

Claim 2a. *Show that $\det : D_n \rightarrow R^*$ has image $+1, -1$.*

Proof. Since $D_n \subset O(n)$, then for all $A \in D_n$ $\det(A)$ is 1 or -1 by the previous homework. Since D_n contains reflections and rotations. By the previous homework we know that the determinant of a rotation is 1 and the determinant of a reflection is -1. Thus $\det^*(D_n) = \{1, -1\}$. $\square$

Claim 2b. *Explain why these are uniquely determined by where they send any fixed vertex, that this must go to another vertex, and that thus there are at most n of them. Conclude there are exactly n , and that $|D_n| = 2n$ and that the element you found in 1) are the whole group.*

Proof. Consider a fixed vertex of V of the regular n -gon. Any rotation in $R \in C_n$ must map this fixed vertex V to another vertex to preserve the shape of polygon. Assume the contrary we have two different rotations R_1, R_2 such that $R_1V = R_2V$. Then $R_1R_2^{-1}V = V$. However the product of R_1R_2 is also a rotation because $SO(2)$ is a group. Moreover it means that R_1R_2 has a eigenvalue of 1 with eigenvector $V \neq 0$. However this only true if $R_1R_2^{-1} = R_0 = I$, Thus $R_1 = R_2^{-1-1} = R_2$ and R_1 and R_2 are the same rotation thus forming a contradiction. Thus it means each rotation in D_n is

uniquely determined on which vertex it sends a vertex V to. Since there are n vertices in the n -gon, thus there are n choices on where to send V to. Thus there are n rotations in D_3 . Furthermore since all rotations form the group $C_n = \ker \det$ and all reflections form the coset $\det^*(-1) = \ker \det + r_{y=0}$ $|C_n| = |\det^*(-1)|$ and thus there are n reflections. Thus $|D_n| = n$ $\square$

Claim 2c. Explain why $C_n \cong \mu_n$, the group of n th roots of unity.

Proof. The vertices of the n -gon are on the n th roots of unity and the n th roots of unity are rotations in an of themselves. Let $f : C_n \rightarrow \mu_n$, f takes $R_{2\pi k/n} \rightarrow e^{2i\pi k/n}$ for $k \in [0, \dots, n-1]$. Let $g : \mu_n \rightarrow C_n$, g takes z where $z = e^{2i\pi k/n}$ for $k \in [0, \dots, n-1]$ to $R_{2\pi k/n}$. $\forall R_{2\pi k/n} \in C_n$ $g(f(R_{2\pi k/n})) = g(e^{2i\pi k/n}) = R_{2\pi k/n}$. For all $e^{2i\pi k/n} \in \mu_n$, $f(g(e^{2i\pi k/n})) = g(R_{2\pi k/n}) = e^{2i\pi k/n}$. Thus f is bijective. $\forall R_{2\pi k/n}, R_{2\pi l/n} \in C_n$, $f(R_{2\pi k/n} R_{2\pi l/n}) = f(R_{2\pi(k+l)/n}) = e^{2i\pi(k+l)/n} = e^{2i\pi k/n} e^{2i\pi l/n} = f(R_{2\pi k/n}) f(R_{2\pi l/n})$. Thus f is a bijective homomorphism and is an isomorphism. Thus $C_n \cong \mu_n$. $\square$

Claim 3a. Let x be a generator of C_n and $y \in D_n$ any reflection. Explain why $\langle x, y \rangle = D_n$, ie D_n is generated by these two elements.

Proof. $C_n = \langle x \rangle \subset D_n$ by definition of x and C_n . Since $C_n \cong \mu_n$, $|C_n| = n$. Furthermore $\langle x, y \rangle \subset D_n$ is a subgroup of D_n and $C_n \subset \langle x, y \rangle$. By Lagrange's Theorem, $|C_n| \mid |\langle x, y \rangle| = n \mid |\langle x, y \rangle|$ or $|\langle x, y \rangle| = kn$ for $k \in \mathbb{Z}$. Furthermore by Lagrange's Theorem, $|\langle x, y \rangle| \mid |D_n| = 2n$, thus $2n = akn$ for $a, k \in \mathbb{N}$ (order has to be positive). $ak = 2$, furthermore $k \neq 1$ because $|\langle x, y \rangle| > |\langle x \rangle|$ since $y \in \langle x, y \rangle$ but $y \notin \langle x \rangle$. Thus $a = 1$ and $k = 2$. Thus $|\langle x, y \rangle| = 2n = |D_n|$, thus since the order is the same and $\langle x, y \rangle$ is a subgroup then $\langle x, y \rangle = D_n$ $\square$

Claim 3b. Show they satisfy the relations $x^n = y^2 = 1$ and $xyx = x^{-1}$.

Proof. x by definition generates C_n . $C_n \cong \mu_n$ by claim 2c. thus $|x| = |m|$ where m is the generator of μ_n , thus $|m| = n$. Thus $|x| = n$ and $x^n = 1$.

y is by definition a reflection. By the previous homework, we know its eigenvalues are $1, -1$. $\forall v \in V_1(y)$, $y^2 v = yyv = yv = v = Iv$, and $\forall w \in V_2(y)$, $y^2 w = y(-w) = w = Iw$. Moreover since y has 2 distinct eigenvalues, and $y \in O(2)$ it must be diagonalizable and have two eigenvector basis v, w s.t $yv = v$ and $yw = -w$ (eigenvalues are 1 and -1). Thus for all $x \in \mathbb{R}^2$, $x = av + bw$, and $y^2 x = y(yx) = y(y(av + bw))$, by linearity $y(y(av + bw)) = y(ayv + byw) = ay^2 v + by^2 w = av + bw = x = Ix$. Thus $y^2 = 1$. By the previous homework problem 7, we know if $x = \rho_\theta$ and y is a reflection $xyx^{-1} = y\rho_\theta y^{-1} = \rho_{-\theta} = \rho_\theta^{-1} = x^{-1}$. Thus $xyx^{-1} = x^{-1}$. $\square$

Claim 3c. Now let G be any group generated by two elements X, Y satisfying these relations. Explain why every element of G can be written as $X^a Y^b$ with $a \in \{0, \dots, n-1\}$ and $b \in \{0, 1\}$.

Proof. Since $G = \langle x, y \mid x^n = y^2 = 1, xyx^{-1} = x^{-1} \rangle$. Since in G , $xyx^{-1} = x^{-1}$ we know $yx = x^{-1}y$. Moreover we know $x^n = x^{n-a}x^a = 1$ thus $x^{-a} = x^{n-a}$ for $a \in [0, \dots, n-1]$. Thus we know $yx = x^{n-1}y$ and equivalently $yx^{n-1} = xy$. However more generally we know, $yx^a = x^{n-1}yx^{a-1} = x^{-1}x^{-(a-1)}y = x^{n-1}x^{n-a+1}y = x^{2n-a}y = x^{-a}y$. Thus $yx^a = x^{-a}y$ and $yx^{-a} = x^a y$. Thus we have found a way to swap x and y neatly by utilizing their relations. Since x and y generate G , $\forall g \in G$, $g = x^{a_1} y^{b_1} \dots x^{a_k} y^{b_k}$. However since $yx^{-a} = x^a y$, we can swap the values of x and y till we get $g = x^p y^q$ where p, q depends on a_i, b_i for all $i \in [1, \dots, k]$ however since $x^n = y^2 = 1$, $g = x^p y^q = x^s y^t$ where $s = p \mod n$ and $t = q \mod 2$. $\square$

Claim 3d. In particular $|G| \leq 2n$. Show that in the case of D_n , this expression is unique, but give an example of a group where it is not (ie where $|G| < 2n$).

Proof. There are only n options for s , $[0, \dots, n-1]$ and only 2 options for t , $\{0, 1\}$. Thus $|G| \leq 2n$. Since there are no additional restrictions placed on x, y then $|D_n| = 2n$. Take $D_2 = \langle x, y \mid x^2 = y^2 = 1, yx = x^{-1}y \rangle = \langle x \mid x^2 = 1 \rangle, |D_2| = 2$. $\square$

Claim 3e. Show that the relations completely determine the multiplication in G , ie show that $X^a Y^b X^c Y^d = X^s Y^t$ for s, t determined by a, b, c, d and the above relations.

Proof. As previously shown, $yx^a = x^{-a}y$ for all $a \in \mathbb{Z}$. Thus $X^a Y^b X^c Y^d = X^a Y^{b-1} X^{-c} Y Y^d = X^a X^{(-1)^b c} Y^b Y^d = X^{a+(-1)^b c} Y^{b+d}$. Thus $X^a Y^b X^c Y^d = X^{a+(-1)^b c} Y^{b+d}$. However since $X^n = 1$ and $Y^2 = 1$, $X^{a+(-1)^b c} = X^s$ where $s = (a + (-1)^b c) \bmod n$ and $Y^{b+d} = Y^t$ where $t = (b + d) \bmod 2$. Thus $X^a Y^b X^c Y^d = X^s Y^t$ and the relations completely determine the multiplication in G . $\square$

Problem 4. Now the goal is to show that, for any group G , $\text{Hom}_{Gr}(D_n, G)$ is naturally the set of pairs of elements $(X, Y) \in G$ which satisfy the above relations. We construct a natural bijection between this Hom set and this set of pairs. The forward direction is easy – just evaluate at x and y . In the other direction let (X, Y) be such a pair.

Explain why there is at most one homo from $D_n \rightarrow G$ sending $x \rightarrow X$ and $y \rightarrow Y$, we want to show one exists. First show that if it does exist then it sends $x^a y^b \rightarrow X^a Y^b$. Show there does exist a unique function which does this. Using 3) show this is a homomorphism.

Claim 4a. If there exists a homomorphism $f : D_n \rightarrow G$, it sends $x^a y^b \rightarrow X^a Y^b$

Proof. Let $f : D_n \rightarrow G$ be a homomorphism. Since f is a homomorphism $f(x^a) = f(x)^a = X^a$ and $f(y^b) = f(y)^b = Y^b$. Moreover since f is a homomorphism, it must preserve the group structure and its relations so $X^n = Y^2 = 1$ and $YX = X^{-1}Y$. Thus since f is homomorphism, $f(x^a y^b) = f(x^a) f(y^b) = X^a Y^b$. $\square$

Claim 4b. There is a unique function that does this from $D_n \rightarrow G$

Proof. Construct a function $f : D_n \rightarrow G$ as the following $f(x) = X$ and $f(y) = Y$ where $(X, Y) \in \{(X, Y) \mid X^n = Y^2 = 1, YX = X^{-1}Y\}$ and that $f(x^a) = X^a$, $f(y^b) = Y^b$, and $f(x^a y^b) = X^a Y^b$. Since $\forall g \in D_n, g = x^s y^t$ for some $s \in [0, \dots, n-1]$ and $t \in [0, 1]$, $f(g) = X^s Y^t$. Thus the function is well defined over every element in D_n .

Suppose there exist 2 functions f, g such that $f(x^a) = f(x)^a = X^a$, $f(y^b) = g(y^b) = Y^b$, and $f(x^a y^b) = g(X^a Y^b) = X^a Y^b$. Then they agree on every value of G and thus must be the same function. Thus there is only 1 unique function $\square$

Claim 4c. The function is a homomorphism

Proof. $f(x^a y^b) f(x^c y^d) = (X^a Y^b)(X^c Y^d) = X^a Y^b X^c Y^d$. Since the relations completely determine the multiplication in G and $(X, Y) \in \{(X, Y) \mid X^n = Y^2 = 1, YX = X^{-1}Y\}$ by definition, by 2d we know $X^a Y^b X^c Y^d = X^s Y^t$ where $s = (a + (-1)^b c) \bmod n$ and $t = (b + d) \bmod 2$. $f(x^a y^b) f(x^c y^d) = X^s Y^t = f(x^s y^t)$, by 3b we know $x^a y^b x^c y^d = x^s y^t$. Thus $f(x^a y^b) f(x^c y^d) = f(x^s y^t) = f(x^a y^b x^c y^d)$. Thus f is a homomorphism. $\square$

Problem 5. What 4) shows is that D_n is "the group given by generators and relations $\langle x, y \mid x^n = y^2 = 1, yxy^{-1} = x^{-1} \rangle$ ". Quotes because it is not such a simple thing to say what it means for a group to be "the group" given by some generators and relations. This uses the notion of universal property, 4) is an example. We will turn to this later.

Problem 6. Consider μ_n (roots of unity), and let x be a generator. Note $x^n = e = 1$. Show this has the analogous universal property, ie $\text{Hom}_{gr}(\mu_n, G)$ is naturally the set of elements $g \in G$ which satisfy $g^n = e$.

Proof. For all $f \in \text{Hom}_{gr}(\mu_n, G)$ and $x \in \mu_n$ s.t $\mu_n = \langle x \rangle$, $f(x) = g \in G$. Since f is by definition a homomorphism, $f(x^n) = f(x)^n = g^n$, however $f(x^n) = f(e) = e$ because $x^n = e$ thus $g^n = e$. Thus for each $f \in \text{Hom}_{gr}(\mu_n, G)$, $\exists g \in G$ s.t $g^n = e$. Furthermore for every $g \in G$ s.t $g^n = e$ there exists a function s.t $[x \rightarrow g]$. Thus the map $\text{Hom}_{gr}(\mu_n, G) \rightarrow \{g \mid g^n = e\}$ is surjective. Furthermore let $f_1, f_2 \in \text{Hom}_{gr}(\mu_n, G)$ such that $f_1(x) = f_2(x) = g$ since they take the generator to the same element and every $y \in \mu_n$ is $y = x^k$ for $k \in \mathbb{Z}$, $f_1(y) = f_2(y) = f_1(x^k) = f_1(x)^k = f_2(x)^k = f_2(x)^k$. Since both functions take every input value to the same output they must be the same function. Thus the map from $\text{Hom}_{gr}(\mu_n, G) \rightarrow \{g \mid g^n = e\}$ is bijective.

Let $f : \text{Hom}_{gr}(\mu_n, G) \rightarrow \{g \mid g^n = e\}$ be the bijective map. It has a inverse map, $g = f^{-1} : \{g \mid g^n = e\} \rightarrow \text{Hom}_{gr}(\mu_n, G)$. Let $a, b \in \{g \mid g^n = e\}$, $g(ab) = [x \rightarrow ab] = [x \rightarrow a][x \rightarrow b] = g(a)g(b)$ thus g is a homomorphism and f must be a homomorphism as well

Thus $\text{Hom}_{gr}(\mu_n, G) \cong \{g \mid g^n = e\}$. □

Problem 7. Even simpler is the group $\mathbb{Z}$ (integers under $+$). Let $X = 1$. Show that $\text{Hom}_{gr}(\mathbb{Z}, G)$ is naturally G – the set of elements of G .

Proof. (Same as group 3 problem 2B)

Let $f : \text{Hom}_{groups}(\mathbb{Z}, G) \rightarrow G$ s.t it maps $\phi \rightarrow \phi(1)$.

Suppose $f(\phi_1) = f(\phi_2)$. Then, $\phi_1(1) = \phi_2(1)$. Since ϕ_1 and ϕ_2 are group homomorphisms acting on the group of integers with addition, thus $\phi(n) = \phi(1^n) = \phi(1)^n$ (not exponentiation is in context of group exponentiation), they are completely determined by their values on 1. Thus, $\phi_1 = \phi_2$, and ϕ is injective.

For any $g \in G$, consider the homomorphism $\phi_g : \mathbb{Z} \rightarrow G$ defined as $\phi_g(n) = g^n$. For all $a, b \in \mathbb{Z}$ $\phi_g(ab) = g^{ab} = g^a g^b = \phi_g(a)\phi_g(b)$. Now, notice that $f(\phi_g) = \phi_g(1) = g^1 = g$. Since we can find such a homomorphism ϕ_g for any $g \in G$, f is surjective.

Since f is surjective and injective, it is bijective. Also since its also a homomorphism it is a isomorphism. □

Proof. Can you find the group $\langle x, y \mid xy = yx \rangle$? So a group, D , such that $\text{Hom}_{gr}(D, G) = \{X, Y \in G \mid XY = YX\}$? We will show later that there exists a unique such group for any set of generators and relations. □

Proof. $\mathbb{Z}^2 = \langle x, y \mid xy = yx \rangle$, because $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ operation is addition. $x+y = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = y+x$ □